BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Compression and Storage Calculations

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

SILENT

BLACK PEN: Answer all four before looking at the model answers.

1. Previous lesson: what does sample rate mean?

2. How many bits make a byte?

3. Calculate bytes: 8,000 samples/sec for 2 sec at 8 bits.

4. Why does doubling recording duration increase file size?

Today's objectives

Compare lossy and lossless methods; calculate storage saved by compression.

Key vocabulary - Read, cover, recall

compression
Reducing the number of bits in a file

lossless
Original data can be restored exactly

lossy
Some original information is discarded

ratio
Comparison of original and new sizes

Write one definition in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Reversible or discarded?

Compression stores a file with fewer bits. The method matters.

LOSSLESS

All original data can

be restored exactly.

LOSSY

Some information is lost

to reduce file size.

Example: programming files need lossless compression.

I Do - Compare file sizes

LISTEN

Original 5000 KB; compressed 3000 KB.



Saving = 5000 - 3000 = 2000 KB

Percentage saved = 2000 / 5000 x 100 = 40%

We Do - Guided practice

MWBs

A 2,000 KB file becomes 1,250 KB. Find KB and percentage saved.

KB saved = _____ - _____ = _____

Fraction saved = _____ / _____

Percentage saved = _____ x 100 = _____%

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

MWBs

Start on paper. Do not look at model answers until page 4.

```
original = 5000    compressed = 3000
print(original - compressed)
```

Predict how many kilobytes the program says were saved.

Write a print instruction to show the percent saved.

Change compressed to 2500. Predict the new saving.

You Do - Independent work

SILENT

State the difference between lossy and lossless. [2]

An image shrinks from 5000 KB to 3000 KB. How much is saved? [2]

Calculate the percentage saved when 5 MB becomes 3 MB. [3]

Thinking: Which compression should be used for source code? Why?

Exit Ticket - True or False?

THUMBS

Answer before you turn to the green-pen page.

1. Lossless compression can reproduce the original file exactly. ☐ T ☐ F
2. Lossy compression always keeps every original detail. ☐ T ☐ F
3. Compressed files can take less storage space. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick correct answers. For a mistake, use the model answer then copy the FULL question and write a fresh answer below.

Do Now - Answers

Q1 Samples taken every second. ☐

Q2 8 bits. ☐

Q3 16000 bytes. ☐

Q4 Twice as many samples are stored. ☐

We Do - Guided practice

750 KB saved; 750 / 2000 x 100 = 37.5%

Key vocabulary

compression ☐
Reducing the number of bits in a file

lossless ☐
Original data can be restored exactly

lossy ☐
Some original information is discarded

ratio ☐
Comparison of original and new sizes

Code First - Answers

GREEN PEN

Predict ☐

5000 - 3000 = 2000 KB

Write ☐

print((original - compressed) / original * 100) gives 40

MWBs

Improve ☐

5000 - 2500 = 2500 KB

Exit Ticket - Explanations

1. True

2. False: lossy discards some data

3. True

You Do - Self-mark

GREEN PEN

Q1	Lossless is reversible; lossy discards information	☐
Q2	5000 - 3000 = 2000 KB saved	☐
Q3	(5 - 3) / 5 x 100 = 40%	☐
Thinking	Lossless: every character must be recoverable exactly.	☐

Try the question again

GREEN PEN

1. Copy the FULL question you got wrong (or the hardest):

2. In GREEN PEN, write a NEW answer. Show working: